# 13.5 Asking Awkward Questions

**WORKSHEET**

**The question**

(i) $X_1$ and $X_2$ are both random variables which take values $x_1, x_2, \ldots, x_n$, with probabilities $a_1, a_2, \ldots, a_n$ and $b_1, b_2, \ldots, b_n$ respectively.

The value of random variable $Y$ is defined to be that of $X_1$ with probability $p$ and that of $X_2$ with probability $q = 1 - p$.

If $X_1$ has mean $\mu_1$ and variance $\sigma_1^2$, and $X_2$ has mean $\mu_2$ and variance $\sigma_2^2$, find the mean of $Y$ and show that the variance of $Y$ is $p\sigma_1^2 + q\sigma_2^2 + pq(\mu_1 - \mu_2)^2$.

(ii) To find the value of random variable $B$, a fair coin is tossed and a fair six-sided die is rolled. If the coin shows heads, then $B = 1$ if the die shows a six and $B = 0$ otherwise; if the coin shows tails, then $B = 1$ if the die does not show a six and $B = 0$ if it does.

The value of $Z_1$ is the sum of $n$ independent values of $B$, where $n$ is large. Show that $Z_1$ is a Binomial random variable with probability of success $\frac{1}{2}$. Using a Normal approximation, show that the probability that $Z_1$ is within 10% of its mean tends to 1 as $n \to \infty$.

(iii) To find the value of random variable $Z_2$, a fair coin is tossed and $n$ fair six-sided dice are rolled, where $n$ is large. If the coin shows heads, then the value of $Z_2$ is the number of dice showing a six; if the coin shows tails, then the value of $Z_2$ is the number of dice not showing a six.

Use part (i) to write down the mean and variance of $Z_2$. Explain why a Normal distribution with this mean and variance will not be a good approximation to the distribution of $Z_2$. Show that the probability that $Z_2$ is within 10% of its mean tends to 0 as $n \to \infty$.

## Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps.

**Ask yourself** $Y$ is $X_1$ with probability $p$ and $X_2$ with probability $q$. What is $\mathrm{E}(Y)$? What is $\mathrm{E}(Y^2)$?

### STAGE 2 Independent randomisation

**Ask yourself** What is $\mathrm{P}(B = 1)$? Are the $n$ values of $B$ independent?

### STAGE 3 One coin for everybody

**Ask yourself** Now one coin toss decides for all $n$ dice. Given heads, what is the distribution of $Z_2$? Given tails? Use part (i).

**Ask yourself** Where is most of the probability? Is there much near the mean $\frac{n}{2}$?

## Try this next

1. **The twins.** (From STEP II, 2005.) Twins share a computer; the one online is Anna with probability $p$, who answers truthfully with probability $a$, or Bella, who does so with probability $b$. I ask them to toss a coin and say whether it came down heads, and get "yes". Show that the probability that it was heads is $\frac{1}{2}$ if and only if $2(ap + bq) = 1$. If I then ask again and get "no", show that the probability of heads is $\frac{1}{2}$; if I get "yes" again, show that it is $\frac{1}{2}$ provided $2(ap + bq) = 1$.

2. **Estimating $\pi$.** In a Warner survey with $p = \frac{3}{4}$, 400 people answer and 160 say yes. Estimate the proportion $\pi$ in group $A$, and find the approximate standard error of your estimate.

3. **Total variance.** Prove the law of total variance for a discrete hidden variable $C$ with several values.

4. **Shared coins.** In part (iii), what is the probability that $Z_2 > \frac{n}{2}$? Explain why, for large $n$, it is close to $\frac{1}{2}$ even though $Z_2$ is almost never close to $\frac{n}{2}$.

## Further reading

► **Stanley Warner, "Randomized response: a survey technique for eliminating evasive answer bias"**, *Journal of the American Statistical Association* 60 (1965).

- **Cynthia Dwork & Aaron Roth, *The Algorithmic Foundations of Differential Privacy*** (free online). The modern theory; the first chapters are accessible.
- **David Spiegelhalter, *The Art of Statistics*** (Pelican), on surveys and their pitfalls.

The full guided solution, and what the question leads to at university, is in Section 13.5:
https://beyondstep.keithmcnulty.org/ch13/awkward-questions.html